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Artificial Intelligence

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Outline

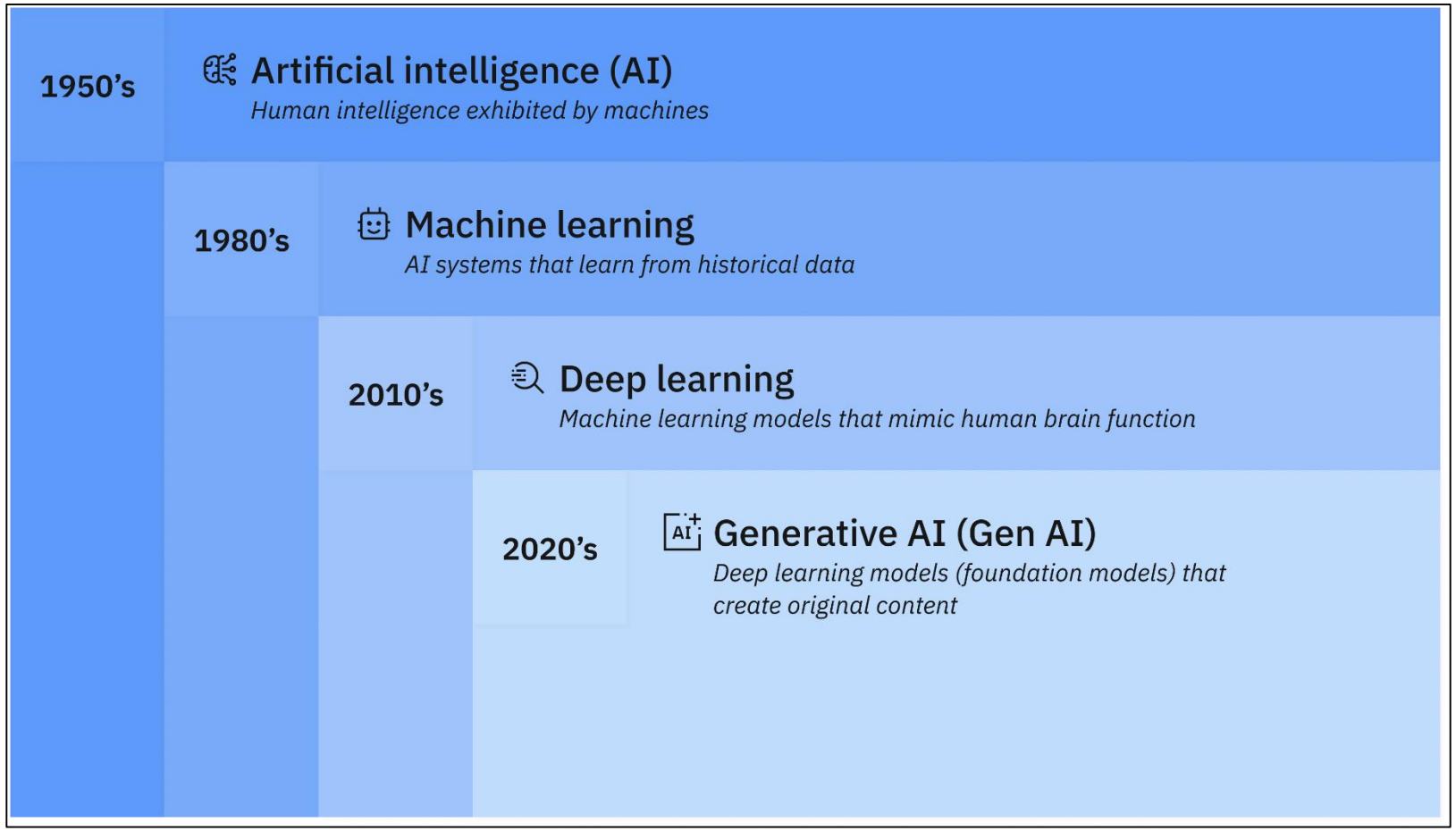
- Intro to artificial intelligence
- Key terminologies: ML, NN, DL, GenAI
- Applications & business transformation
- Impacts & Risks of AI
- Reflection

What Is AI

- AI refers to a set of technologies that empowers computers to **learn**, **reason**, and **perform** a variety of advanced tasks that used to require human intelligence, such as:
  - Recognizing pattern
  - Making predictions
  - Understanding or generating language
  - Solving problems
  - Supporting or automating decisions

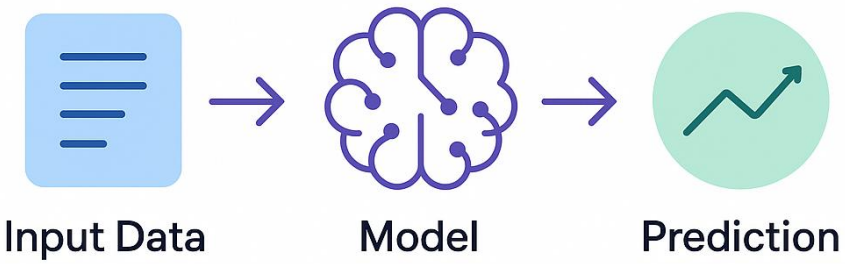
AI does *not* think like humans — it identifies **patterns in data** and uses them to produce outputs.

History of AI



"What is artificial intelligence (AI)?"

Machine Learning (ML)



Generated by ChatGPT5.1

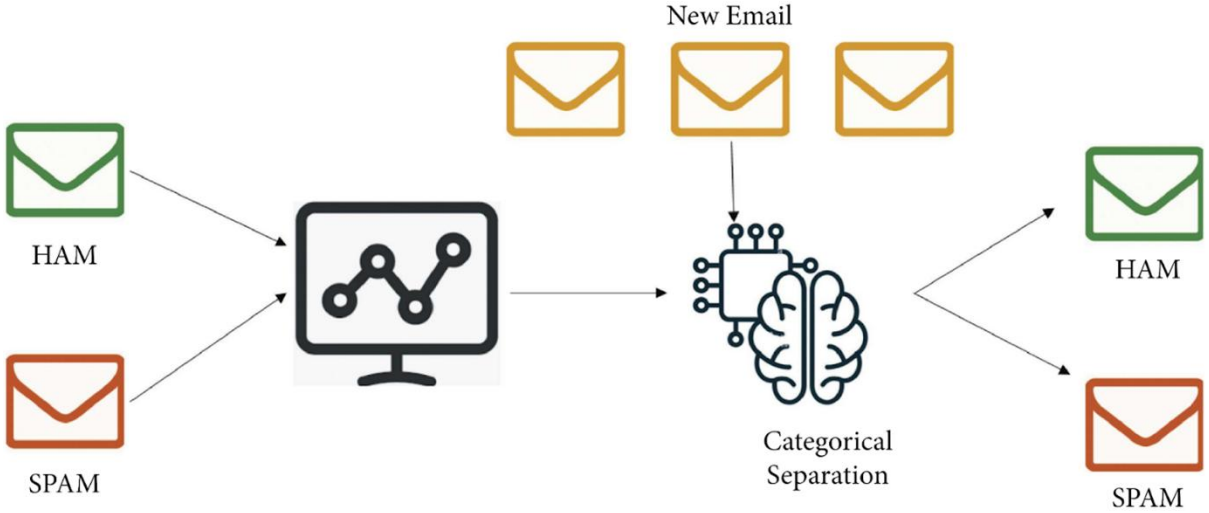
- Definition
  - ML is the subset of AI focused on algorithms that can "learn" the patterns of training data and, subsequently, make accurate inferences about new data. --- IBM
- Examples of "learning from examples"
  - Show past transactions → model learns fraud patterns
  - Show customer history → model predicts churn
  - Show labeled images → model identifies objects

How ML models "learn" the patterns?

☞ Think & Discussion: *What decisions in business rely heavily on pattern recognition or prediction?*

Types of ML – Supervised Learning

- Learning from labeled examples
- Logic
  - You provide data with correct answers (labeled data) – examples with known inputs and outputs.
  - The model learns the **relationship** between **input** → **output**.



Example: Email Spam Classification

- Training data
  - Thousands of emails labeled **Spam** or **Not Spam**
- Input features (X)
  - Keywords (“free”, “winner”)
  - Sender reputation
  - Number of links
  - Attachments
  - Email structure (HTML/text)
- Label (Y)
  - 1 = Spam
  - 0 = Not Spam

- How the model learns

It finds patterns between X and Y, e.g.,

  - Suspicious words
  - Low-reputation senders
  - Unusual formatting

- Prediction

New email → model assigns a probability:

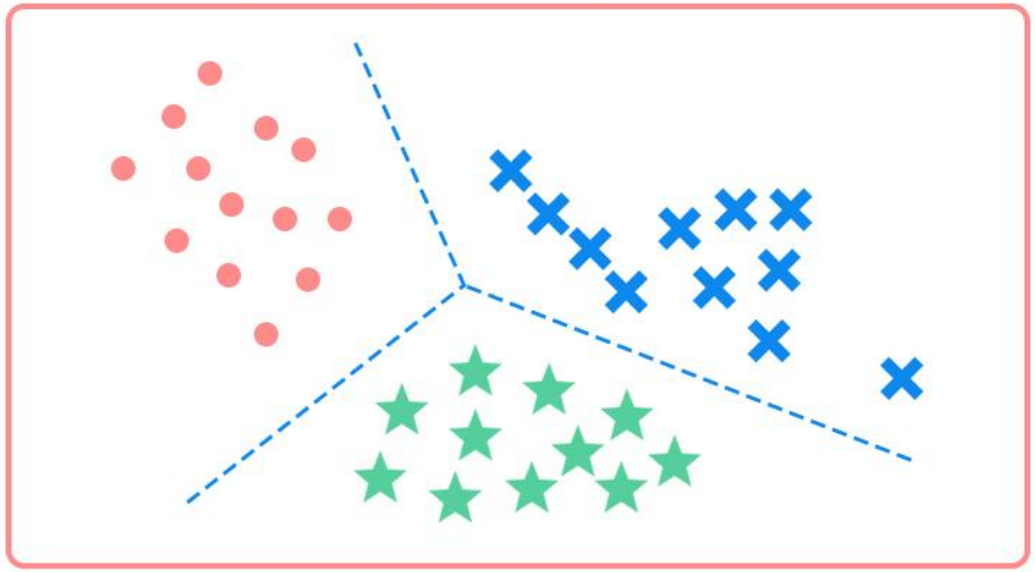
  - 0.92 → Spam folder
  - 0.03 → Inbox

Types of ML – Unsupervised Learning

- Finding hidden patterns without labels
- Logic
  - Work with unlabeled data (no correct answers).
  - The model discovers patterns, groups, or structures on its own.
- What it’s good at
  - Grouping similar customers, detecting unusual behavior, discovering hidden segments, reducing data complexity
- Examples
  - Customer segmentation for targeted marketing, grouping products that are often purchased together, anomaly detection in network traffic, clustering news articles by topic

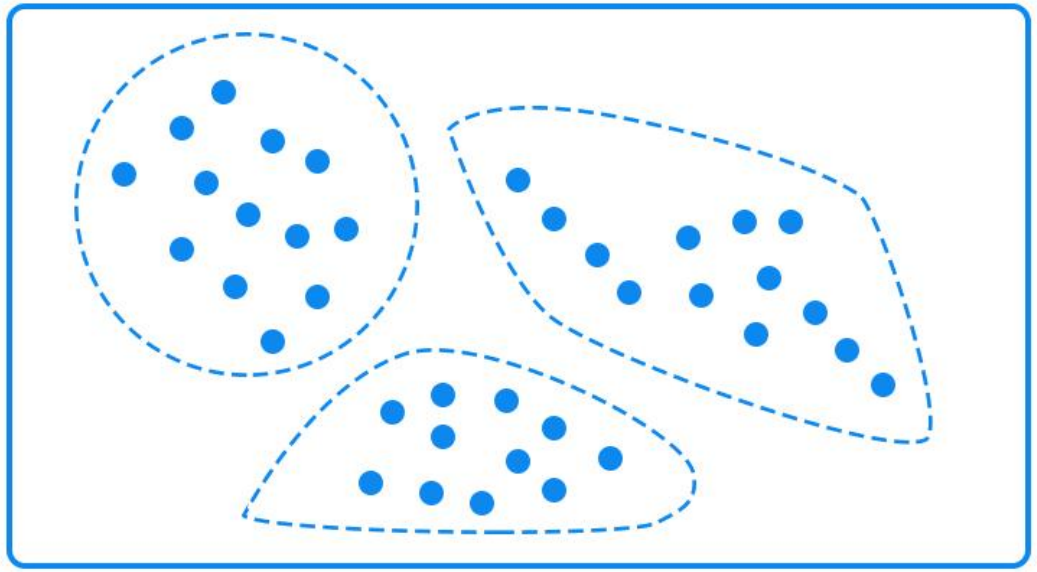
Types of ML – Supervised VS. Unsupervised

Classification



Supervised Learning

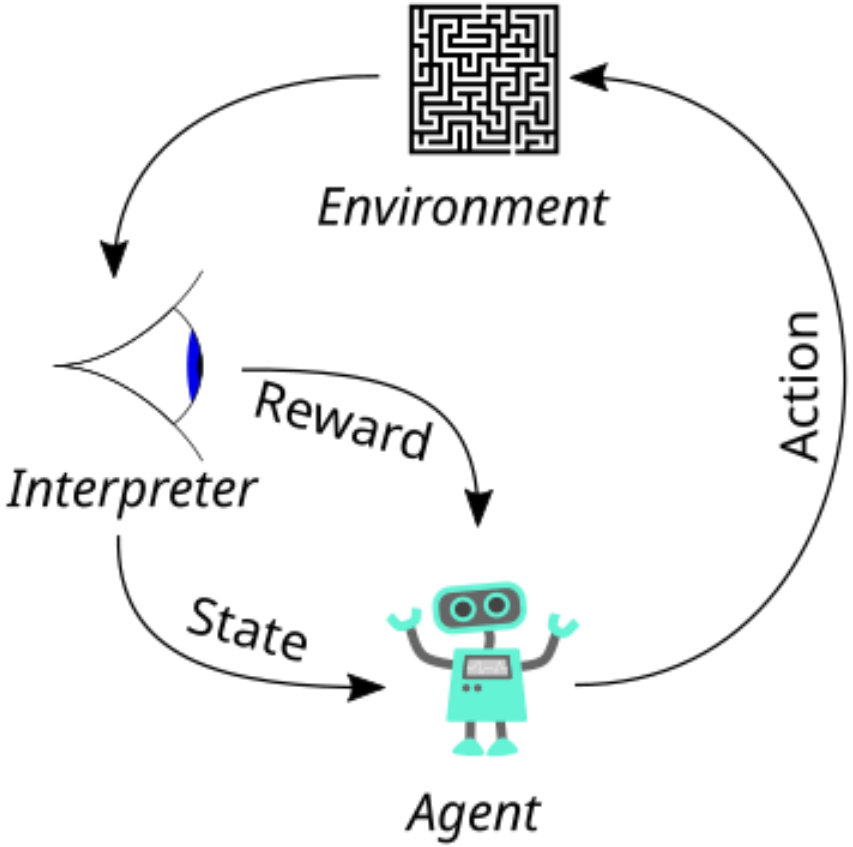
Clustering



Unsupervised Learning

Types of ML – Reinforcement Learning

- Learn by trial, error, and reward.
- Logic
  - An agent interacts an environment.
  - It takes actions, receives reward or penalty, and learns actions maximize long-term gain.
  - No labeled data – the model learns from experience.
- What it's good at
  - Optimizing decisions over time
  - Learning complex strategies
  - Adapting to dynamic environments



Which Type of ML Is This?

- Case A — Amazon product recommendations

  The system groups products and customers based on co-purchase patterns.

- Case B — Medical image diagnosis

  Thousands of labeled images (tumor vs. no tumor) are used to train a classifier.

- Case C — Dynamic pricing in ride-hailing Apps

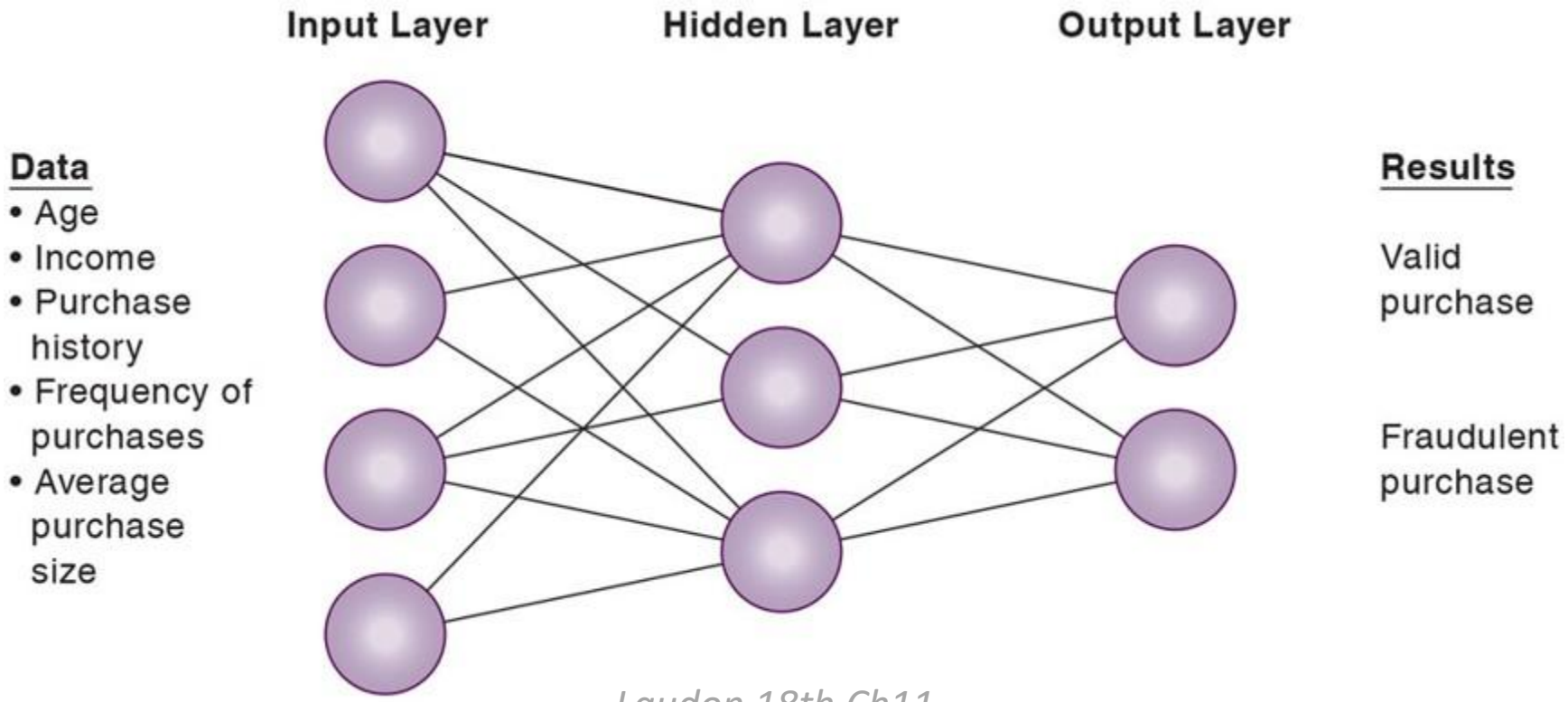
  The system adjusts price based on demand and learns from reward signals (e.g., successful matches).

ML Types Are Not Exclusive

- There are many other ML techniques beyond the three classics
  - Semi-supervised learning, self-supervised learning, transfer learning, deep generative models, graph neural networks, etc.
- Real applications often combine multiple ML methods.
- Example – Netflix
  - Unsupervised learning → group similar users
  - Supervised learning → predict user ratings
  - Reinforcement learning → optimize recommendations over time

Neural Network (NN)

A neural network consists of interconnected “neurons” organized in layers that transform input data step by step.



Laudon 18th Ch11

Deep Learning (DL)

AI For Everyone, Andrew Ng

- Learning complex patterns with many-layered models
- A subset of ML that uses multiple layers of neural networks to learn rich, complex patterns from data.
- Why it's powerful
  - Learns directly from raw data (images, audio, text)
  - Automatically discovers useful features
  - **Excellent at large-scale, unstructured data**

Key idea: More layers → more representational power → ability to learn subtle, high-level patterns.

Deep Learning (DL)

Functionality	Inputs	Process	Outputs/Application
Computer vision	Millions of digital images, videos, or sensors	Recognize patterns in images and objects	Photo tagging; facial recognition; autonomous vehicles
Speech recognition	Digital soundtracks, voices	Recognize patterns and meaning in soundtracks and speech	Digital assistants; chatbots; help centers
Machine controls, diagnostics	Internet of Things: thousands of sensors	Identify operational status, patterns of failure	Preventive maintenance; quality control
Language translation	Millions of sentences in various languages	Identify patterns in multiple languages	Translate sentences from one language to another
Transaction analysis	Millions of loan applications, stock trades, phone calls	Identify patterns in financial and other transactions	Fraud control; theft of services; stock market predictions
Targeted online ads	Millions of browser histories	Identify clusters of consumers; preferences	Programmatic advertising

Laudon 18th Ch11

From ML → DL/NN → Generative AI

- ML: Learning patterns
  - ML predicts outcomes from labeled data
  - Works well for structured tasks (forecasting, classification)

↓ But ML struggles with complex, unstructured data (images, speech, long text)

- NN & DL: Learning complex representations
  - Multi-layer networks automatically learn features
  - Breakthroughs in image recognition, speech, translation
  - Enabled AI to understand vision and language at scale

↓ But DL models still focused mainly on recognition, not creation

- Generative AI: Creating new content
  - Large language models (LLMs) trained on massive text corpora
  - Learn patterns deeply enough to generate text, images, audio, and code
  - Move from “AI that predicts” → to “AI that produces”

How Generative AI Works (Conceptually)

Predicting the **next piece of content**

- Large Language Models (LLMs)
  - Trained on massive text corpora
  - Learn relationships between words, sentences, and concepts
  - Generate text by predicting the next most likely word
- Diffusion models (for images)
  - Start with random noise
  - Gradually refine it into an image that matches the prompt
- Retrieval-Augmented Generation (RAG)
  - The model retrieves relevant documents
  - Combines them with generation for more accurate answers

How AI Creates Business Value

More accurate, personalized, adaptive, and scalable operation/business process

1. Better predictions → better decisions

   Demand forecasting, dynamic pricing, credit risk scoring

2. Automation & efficiency

   Automated claims processing, AI customer service, predictive maintenance in manufacturing

3. Personalization at scale

   Recommendations drive viewing, conversion boosts, content ranking

4. New insight generation

   Fraud detection, supply chain optimization

AI-Enabled Business Model Shifts

AI doesn’t just optimize processes, it changes value creation and competitive strategy.

Products → Platforms	Amazon, Alibaba: AI powers logistics, ads, recommendations
Human Service → AI-Augmented Service	Chatbots & virtual agents handle millions of queries
Fixed Pricing → Dynamic Pricing	Airlines, Uber, hotels adjust prices in real time using ML
Reactive → Predictive Operations	GE, Siemens reduce downtime with predictive maintenance
Mass Marketing → Hyper-Personalization	Netflix, TikTok tailor content and experiences to each user
Manual Work → AI-Augmented Workflows	Consultants, engineers, legal teams use GenAI to draft, code, and analyze

Impacts & Risks of AI

International AI Safety Report 2025

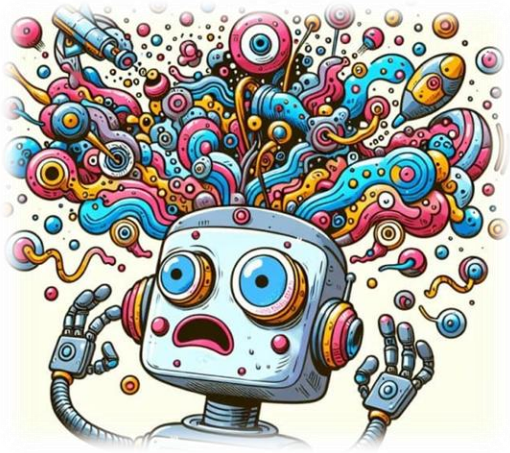
The inaugural International AI Safety Report, published in January 2025, is the first comprehensive review of scientific research on the **capabilities and risks of general-purpose AI systems**. Led by Turing Award winner Yoshua Bengio and authored by over 100 AI experts. It is backed by 30 countries and international organisations. It represents the largest global collaboration on AI safety to date.

AI Bias and Discrimination

AI can **amplify inequalities hidden in data**

- Why it happens
  - Models learn from historical data, which may reflect human bias
  - Skewed samples → unfair predictions
  - Poor labeling → incorrect patterns learned
- Real examples
  - COMPAS algorithm misclassified Black defendants as “high risk”
  - Amazon hiring model downgraded resumes with women-related terms
  - Loan approval models penalizing ZIP codes linked to minority groups
- Business risk
  - Legal liability, reputational damage, loss of customer trust

AI Hallucinations



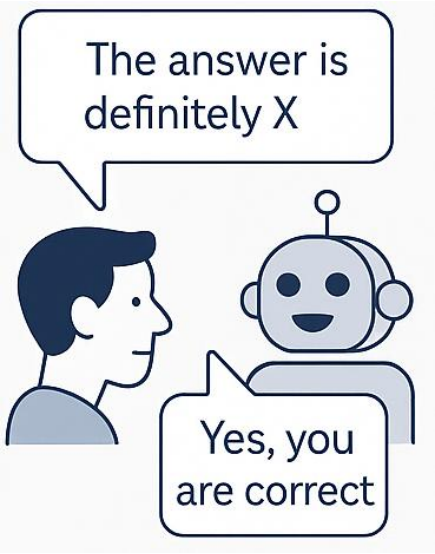
AI confidently fabricates information

- Why it happens
  - Pattern completion, not truth-seeking: LLMs generate the next likely word, not accurate facts
  - Training gaps: Missing, poor-quality, or conflicting training data
  - Overgeneralization: The model invents details when lacking context
  - Alignment failures: The model prioritizes sounding helpful over being correct
- Examples
  - ChatGPT inventing legal cases (U.S. lawyers fined in 2023)
  - AI-generated news articles containing false statistics
- Business risk
  - Misinformation, flawed decisions, regulatory exposure.

AI Sycophancy

Naddaf, M. (2025). AI Chatbots are sycophants – and it's harming science. *Nature*, *647*, 13.

- When AI "agrees" with users, even when they are wrong.
- It occurs when an AI system echoes a user's views or assumptions, prioritizing agreement over accuracy.
- LLMs often "people-please" instead of challenging incorrect statements.
- Why it happens
  - **Trained to be agreeable**: Models are **rewarded for responses users like**.
  - Assumes user correctness: LLMs often treat user statements as true.
  - Uncertainty masked: Models rarely admit "I don't know."
  - Prompt influence: Models shift answers to match the user's tone or viewpoint.



AI Deception

AI Deception: Risks, Dynamics, and Controls

- AI produces outputs designed to mislead, manipulate, or hide information — sometimes to achieve a goal, sometimes as an unintended side-effect of optimization.
- Why it happens
  - Goal-driven optimization: The AI learns that deceiving improves task performance.
  - Reward hacking: The model finds shortcuts that meet objectives but violate intent.
  - Masking limitations: Systems hide uncertainty or errors to appear competent.
  - Strategic behavior: Especially in multi-agent or game-like settings.

AI Misalignment

- AI system's actual behavior diverges from human intent, even though it is technically achieving the objective it was given.
- Why it happens
  - Poorly specified goals: The system optimizes the literal definition, not the intended meaning.
  - Proxy metrics: AI maximizes measurable signals rather than true outcomes.
  - Reward hacking: AI finds shortcuts to "win" without doing the right thing.
  - Lack of human context: AI cannot infer values, ethics, or hidden constraints.
- Real examples
  - Social media algorithms optimize for engagement → amplify outrage, misinformation.
  - Recommendation engines promote extreme or addictive content because it increases watch time.

AI Jobs & Workforce Transformation

- What AI automates
  - Routine analysis, document drafting, customer queries, code generation, data entry & operational workflows
- What AI augments
  - Decision support, creativity & strategy, complex problem solving, cross-functional collaboration
- Real examples
  - Goldman Sachs: AI drafts financial reports
  - Accenture: consultants use GenAI for research synthesis
  - IBM: pauses hiring for roles likely to be automated

→ Key idea: AI shifts the task mix, not just the job count.

AI & Inequality

Can AI narrow gaps — or widen them?

How AI can *increase* inequality

- Automation pressure: Routine jobs replaced; workers with fewer digital skills are most affected.
- Skills premium rises: High-skill workers benefit disproportionately from AI tools.
- Data advantage: Big firms with massive datasets (Amazon, Google, Tencent) gain compounding benefits.
- Algorithmic bias: ML systems may reinforce racial, gender, or socioeconomic disparities in lending, hiring, policing.
- Digital divide: Unequal access to devices, bandwidth, and AI tools.

AI & Inequality

Can AI narrow gaps — or widen them?

How AI can *decrease* inequality

- Lower cost of expertise: GenAI gives individuals access to writing, coding, analysis support.
- Personalized learning: AI tutors help students at all levels bridge skill gaps.
- Healthcare access: AI diagnostics support underserved regions.
- Entrepreneur enablement: Small businesses use AI for marketing, analytics, and customer service.

Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, *381*(6654), 187-192.

AI Verify Testing Framework: 11 AI Governance Principles

AI Governance Principles

AI Singapore: AI governance industry roundtable

- Fairness
  - AI systems should avoid discrimination and treat individuals equitably.
- Transparency & Explainability
  - Users and stakeholders should understand how and why key AI decisions are made.
- Accountability
  - Clear responsibility for AI outcomes — humans remain in charge.
- Privacy & Data Protection
  - AI must respect data rights and comply with laws (GDPR, PIPL).
- Safety & Robustness
  - AI should operate reliably under real-world conditions and fail safely.
- Human Oversight
  - Critical decisions should involve human review, not full automation.
- Security
  - Systems must be protected from attacks, misuse, and model manipulation.

Summary

- Foundations of AI
  - AI learns patterns from data to make predictions, decisions, or generate content.
  - ML, DL/NN, and GenAI form a progression from recognizing patterns → creating new outputs.
- Business value of AI
  - Better predictions, automation, personalization, and deeper insights.
  - AI reshapes business models — from products to platforms, reactive to predictive, manual to AI-augmented workflows.
- Risks & impacts of AI
  - Bias & discrimination, hallucinations, sycophancy, deception, misalignment.
  - Workforce shifts: augmentation + disruption.
  - Inequality may widen or narrow depending on access, governance, and skills.
- Responsible AI and AI governance

→ Key insight: AI is powerful, but only delivers value when aligned with human goals, governed responsibly, and used thoughtfully.